

Engineering Investigation Report

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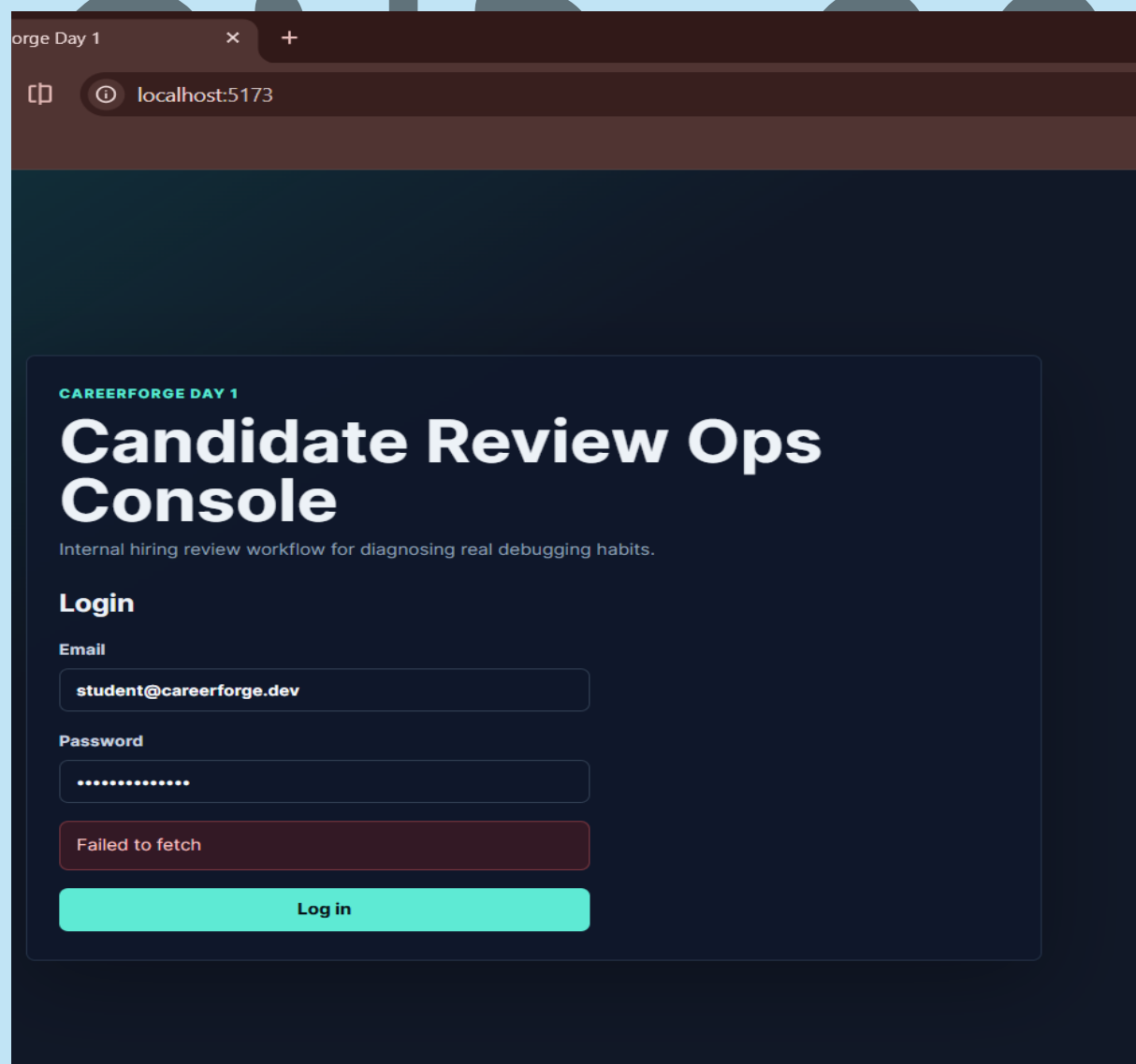
Branch / Year: CO5KC

Repo : https://github.com/gourikpatil291-droid/careerforge7days_challenge_gnp/tree/Day-01-reality-check

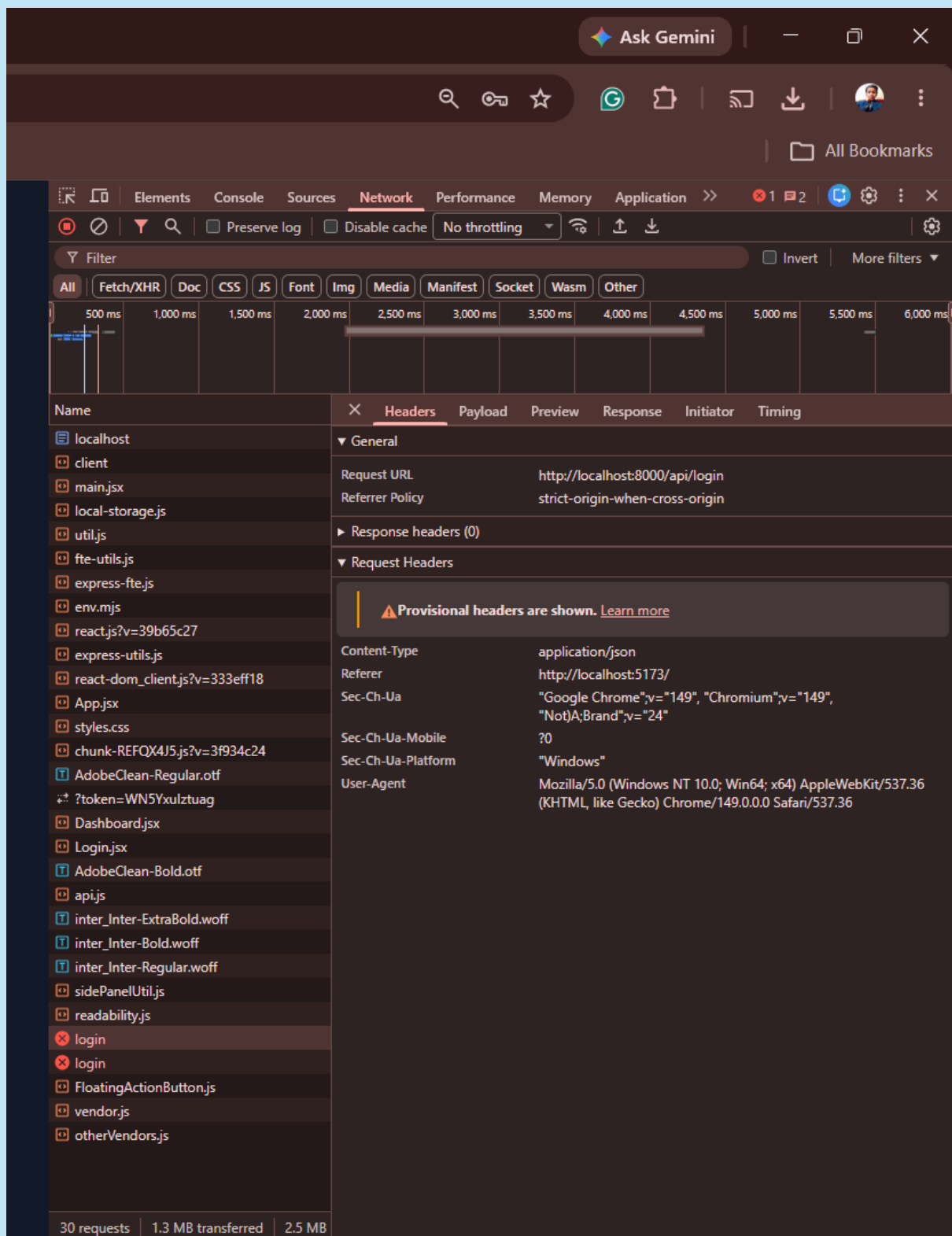
Branch: day-1-reality-check

Evidence :

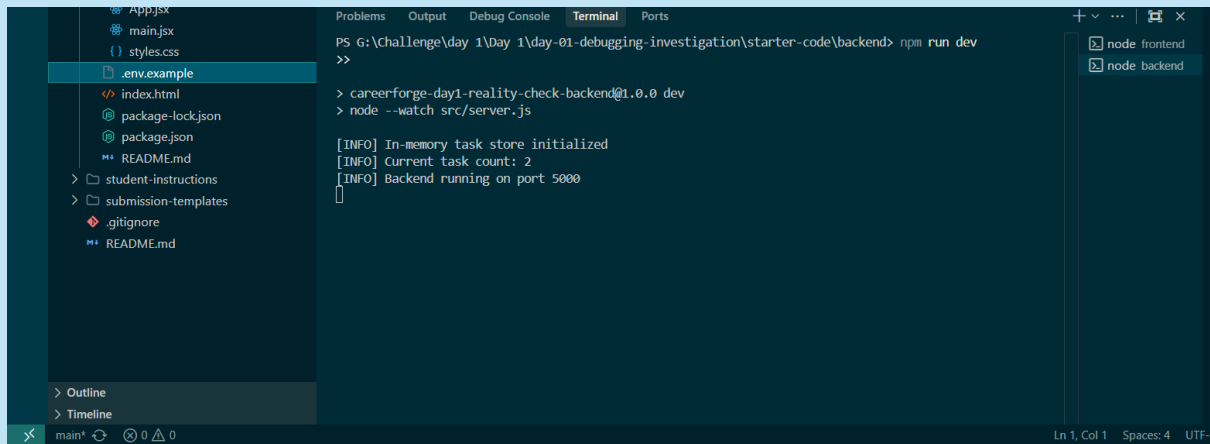
Evidence 1 [E1]



Evidence 2 [E2]



Evidence 3 [E3]

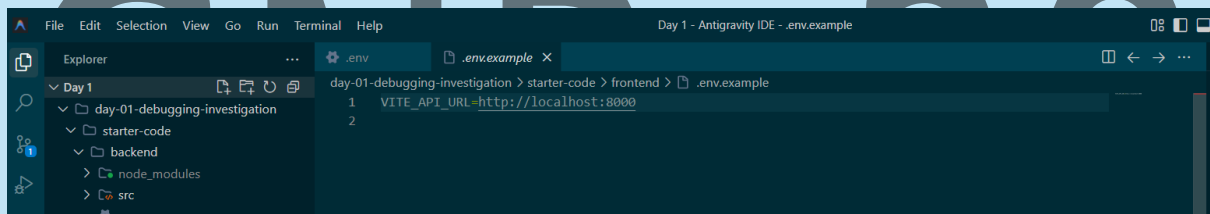


The screenshot shows a VS Code editor with a file explorer on the left displaying a project structure including `App.jsx`, `main.jsx`, `styles.css`, `.env.example`, `package-lock.json`, `package.json`, `README.md`, `student-instructions`, `submission-templates`, `.gitignore`, and `README.md`. The terminal window on the right shows the command `npm run dev` being executed, resulting in the following output:

```
PS G:\Challenge\day 1\Day 1\day-01-debugging-investigation\starter-code\backend> npm run dev
>>
> careerforge-day1-reality-check-backend@1.0.0 dev
> node --watch src/server.js

[INFO] In-memory task store initialized
[INFO] current task count: 2
[INFO] Backend running on port 5000
```

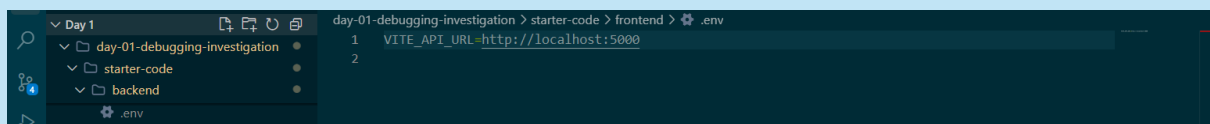
Evidence 4 [E4.1]



The screenshot shows a VS Code editor with a file explorer on the left displaying a project structure including `Day 1`, `day-01-debugging-investigation`, `starter-code`, `backend`, `node_modules`, `src`, and `.env`. The `.env` file is open in the editor, showing the following content:

```
1 VITE_API_URL=http://localhost:8000
2
```

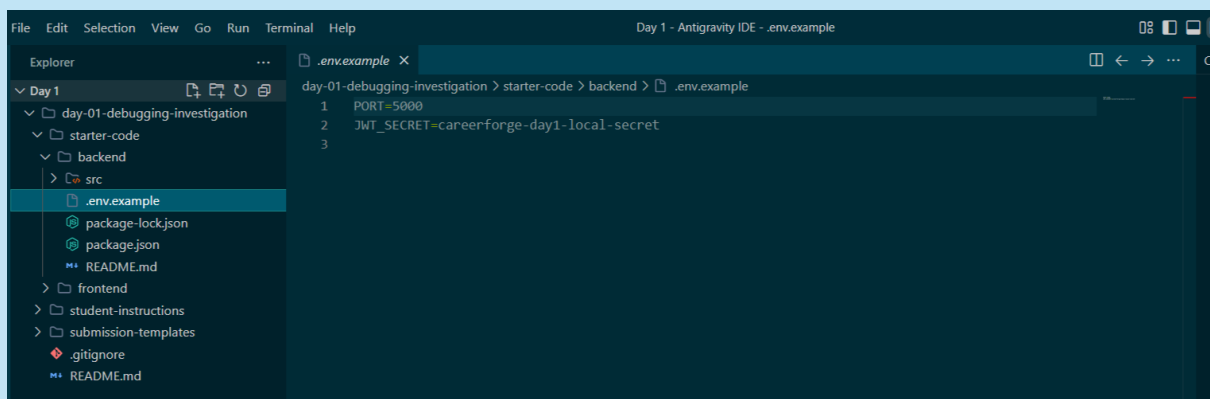
Evidence 5 [E5]



The screenshot shows a VS Code editor with a file explorer on the left displaying a project structure including `Day 1`, `day-01-debugging-investigation`, `starter-code`, `backend`, and `.env`. The `.env` file is open in the editor, showing the following content:

```
1 VITE_API_URL=http://localhost:5000
2
```

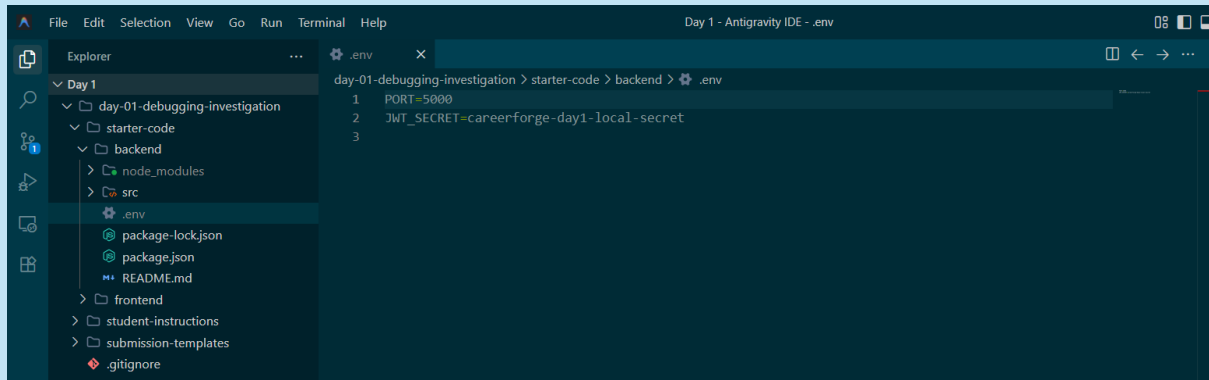
Evidence 5.1 [E5.1]



The screenshot shows a VS Code editor with a file explorer on the left displaying a project structure including `Day 1`, `day-01-debugging-investigation`, `starter-code`, `backend`, `src`, `.env.example`, `package-lock.json`, `package.json`, `README.md`, `frontend`, `student-instructions`, `submission-templates`, `.gitignore`, and `README.md`. The `.env.example` file is open in the editor, showing the following content:

```
1 PORT=5000
2 JWT_SECRET=careerforge-day1-local-secret
3
```

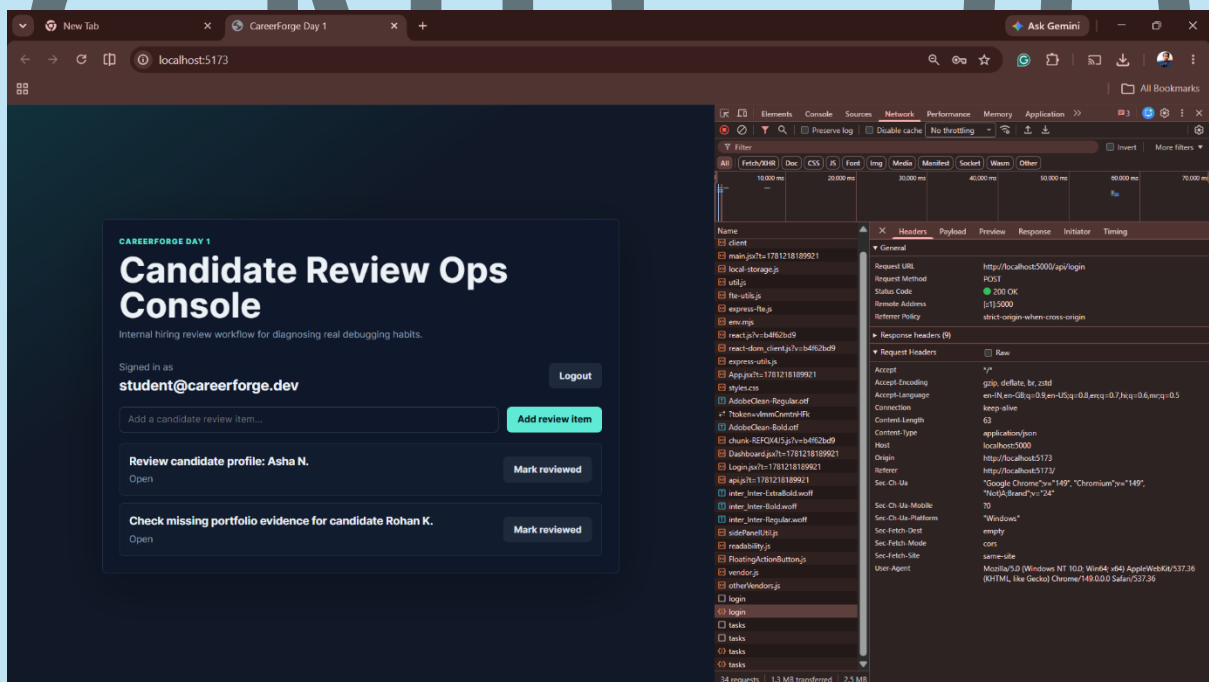
Evidence 5.2 [E5.2]



The screenshot shows the Antigravity IDE interface. The Explorer panel on the left displays the project structure for 'Day 1', including folders for 'day-01-debugging-investigation', 'starter-code', and 'backend', along with files like 'node_modules', 'src', '.env', 'package-lock.json', 'package.json', 'README.md', 'frontend', 'student-instructions', 'submission-templates', and '.gitignore'. The main editor area shows the contents of the '.env' file, which is located at 'day-01-debugging-investigation > starter-code > backend > .env'. The file contains the following configuration:

```
1 PORT=5000
2 JWT_SECRET=careerforge-day1-local-secret
3
```

Evidence 1.1 [E1.1]



The screenshot shows a web browser window with the URL 'localhost:5173'. The page displays the 'CAREERFORGE DAY 1 Candidate Review Ops Console'. The console is signed in as 'student@careerforge.dev' and shows a list of candidate review items. The first item is 'Review candidate profile: Asha N.' with a 'Mark reviewed' button. The second item is 'Check missing portfolio evidence for candidate Rohan K.' with a 'Mark reviewed' button. The right sidebar shows the network tab, displaying a list of requests and a detailed view of the 'POST http://localhost:5000/api/login' request. The request headers include 'Accept: */*', 'Accept-Encoding: gzip, deflate, br, zstd', 'Accept-Language: en-IN,en-GB;q=0.9,en-US;q=0.8,en;q=0.7,hi;q=0.6,ms;q=0.5', 'Content-Length: 63', 'Content-Type: application/json', 'Host: localhost:5000', 'Origin: http://localhost:5173', 'Referer: http://localhost:5173/', 'Sec-Ch-Ua: \"Google Chrome\";v=149\", \"Chromium\";v=149\", \"Not(A;Brand\";v=24\"', 'Sec-Ch-Ua-Mobile: 0', 'Sec-Ch-Ua-Platform: \"Windows\"', 'Sec-Fetch-Dest: empty', 'Sec-Fetch-Mode: cors', 'Sec-Fetch-Site: same-site', and 'User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36'.

Evidence 6 [E6]

The screenshot shows a web browser at localhost:5173 displaying the 'Candidate Review Ops Console' for 'CAREERFORGE DAY 1'. The user is logged in as 'student@careerforge.dev'. The console shows a message: 'Something went wrong while updating the review item.' Below this, there are two review items: 'Review candidate profile: Asha N.' and 'Check missing portfolio evidence for candidate Rohan K.', both with 'Mark reviewed' buttons. To the right, the browser's developer tools are open, showing the Network tab. A request to 'http://localhost:5000/api/tasks/1' is selected, which returned a 401 Unauthorized status. The response headers show 'strict-origin: when: cross-origin'.

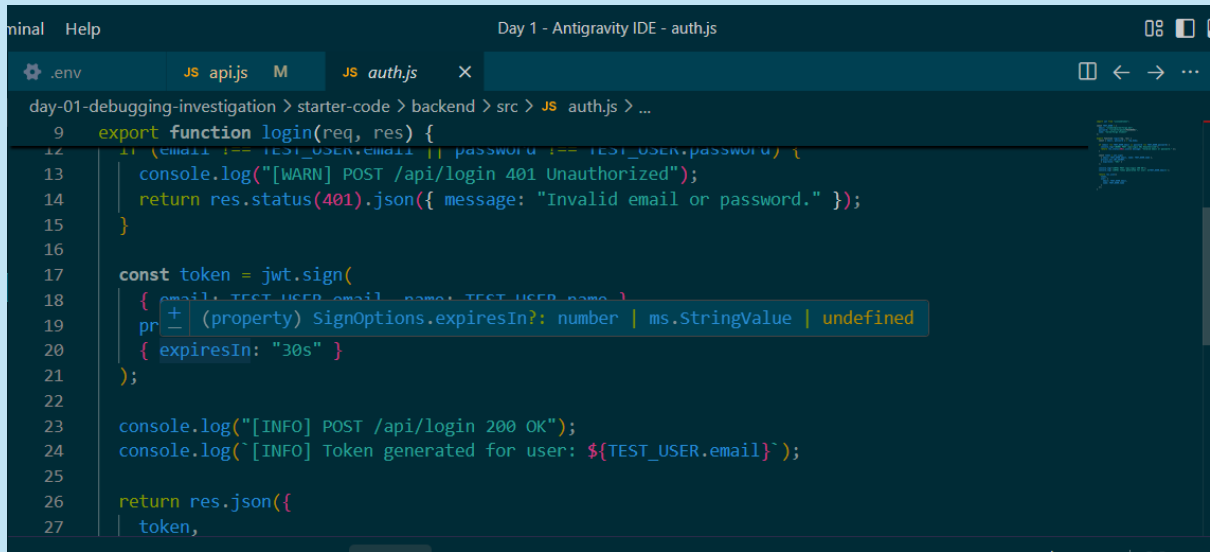
Evidence 1 [E1]

The screenshot shows a code editor with a file explorer on the left showing a project structure with files like 'App.jsx', 'main.jsx', 'styles.css', 'package-lock.json', and 'package.json'. The main editor area shows a JavaScript file with a function that throws an error if the response is not OK. Below the editor, a terminal window shows the command 'npm run dev' being executed. The output of the terminal shows the following logs:

```
PS G:\challenge\day 1\Day 1\day-01-debugging-investigation\starter-code\backend> npm run dev
>>
> careerforge-day1-reality-check-backend@1.0.0 dev
> node --watch src/server.js

[INFO] In-memory task store initialized
[INFO] Current task count: 2
[INFO] Backend running on port 5000
[INFO] POST /api/login 200 OK
[INFO] Token generated for user: student@careerforge.dev
[INFO] GET /api/tasks 200 OK
[INFO] GET /api/tasks 304 Not Modified
[WARN] PATCH /api/tasks/1 401 Unauthorized
[ERROR] TokenExpiredError: jwt expired
```

Evidence 6.2 [E6.2]

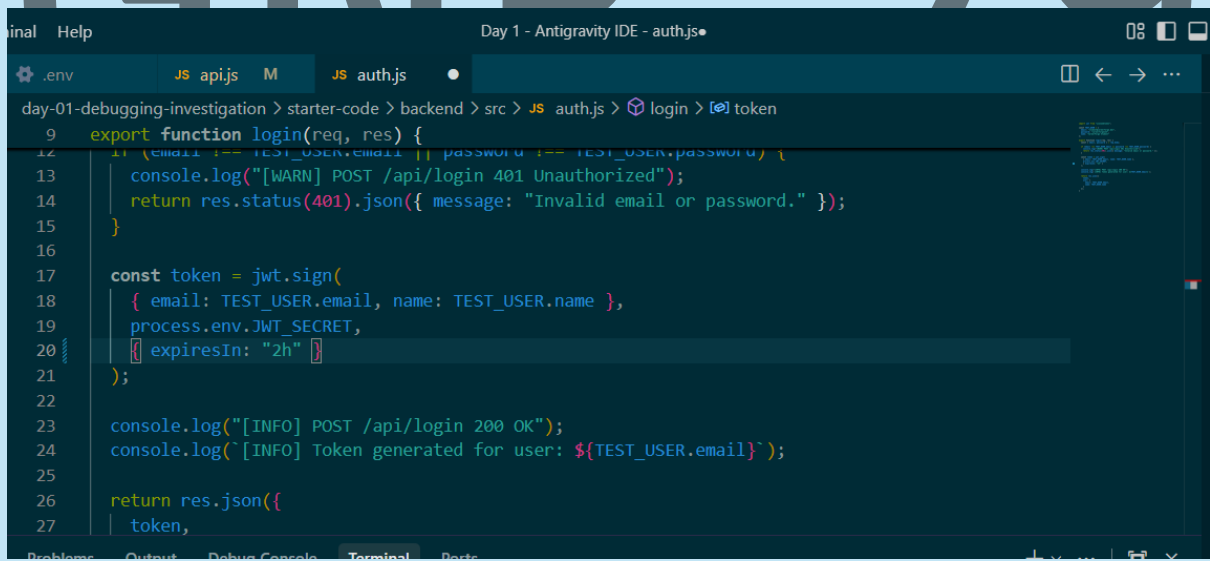


```
Day 1 - Antigravity IDE - auth.js

day-01-debugging-investigation > starter-code > backend > src > JS auth.js > ...

9 export function login(req, res) {
12   if (email !== TEST_USER.email || password !== TEST_USER.password) {
13     console.log("[WARN] POST /api/login 401 Unauthorized");
14     return res.status(401).json({ message: "Invalid email or password." });
15   }
16
17   const token = jwt.sign(
18     { email: TEST_USER.email, name: TEST_USER.name },
19     process.env.JWT_SECRET,
20     { expiresIn: "30s" }
21   );
22
23   console.log("[INFO] POST /api/login 200 OK");
24   console.log(`[INFO] Token generated for user: ${TEST_USER.email}`);
25
26   return res.json({
27     token,
```

Evidence 6.3 [E6.3]

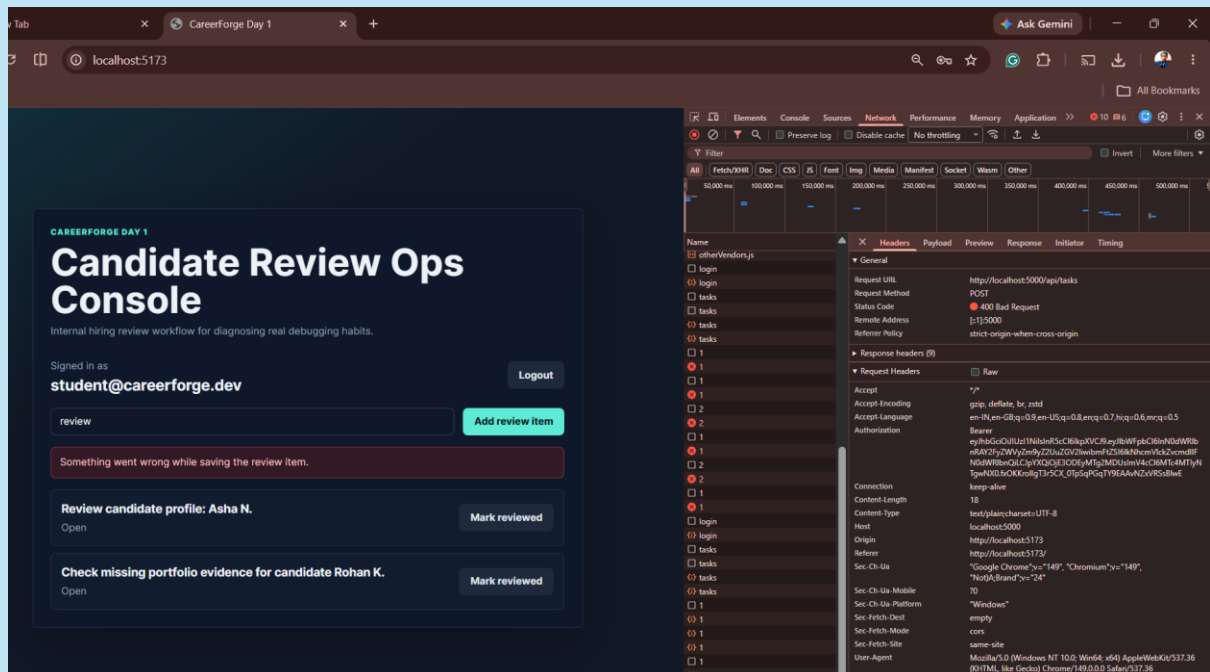


```
Day 1 - Antigravity IDE - auth.js

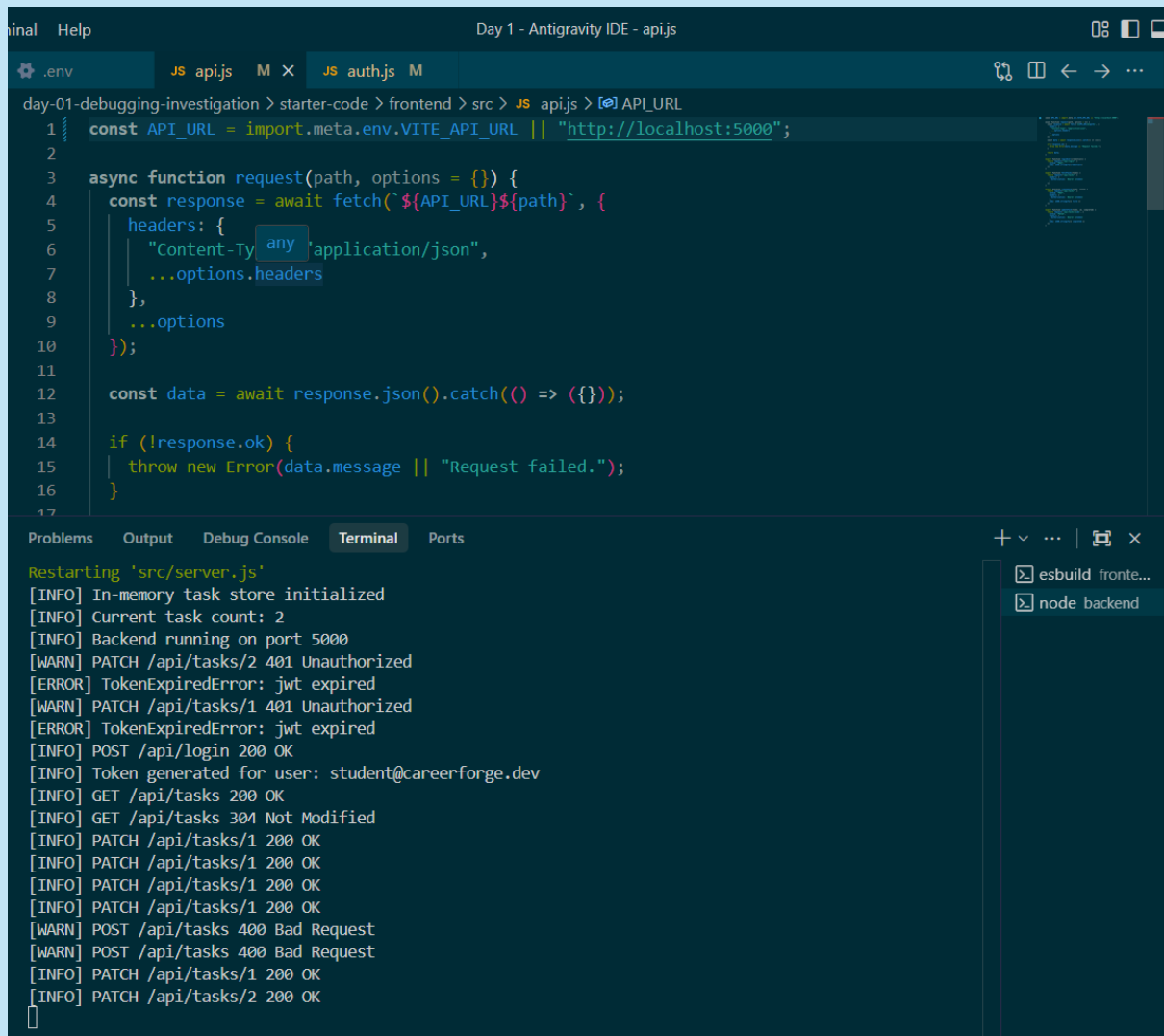
day-01-debugging-investigation > starter-code > backend > src > JS auth.js > login > token

9 export function login(req, res) {
12   if (email !== TEST_USER.email || password !== TEST_USER.password) {
13     console.log("[WARN] POST /api/login 401 Unauthorized");
14     return res.status(401).json({ message: "Invalid email or password." });
15   }
16
17   const token = jwt.sign(
18     { email: TEST_USER.email, name: TEST_USER.name },
19     process.env.JWT_SECRET,
20     { expiresIn: "2h" }
21   );
22
23   console.log("[INFO] POST /api/login 200 OK");
24   console.log(`[INFO] Token generated for user: ${TEST_USER.email}`);
25
26   return res.json({
27     token,
```

Evidence 7 [E7]



Evidence 7.1 [E7.1]



The screenshot displays the Antigravity IDE interface. The top bar shows the file path: `day-01-debugging-investigation > starter-code > frontend > src > JS api.js > API_URL`. The editor shows the following JavaScript code:

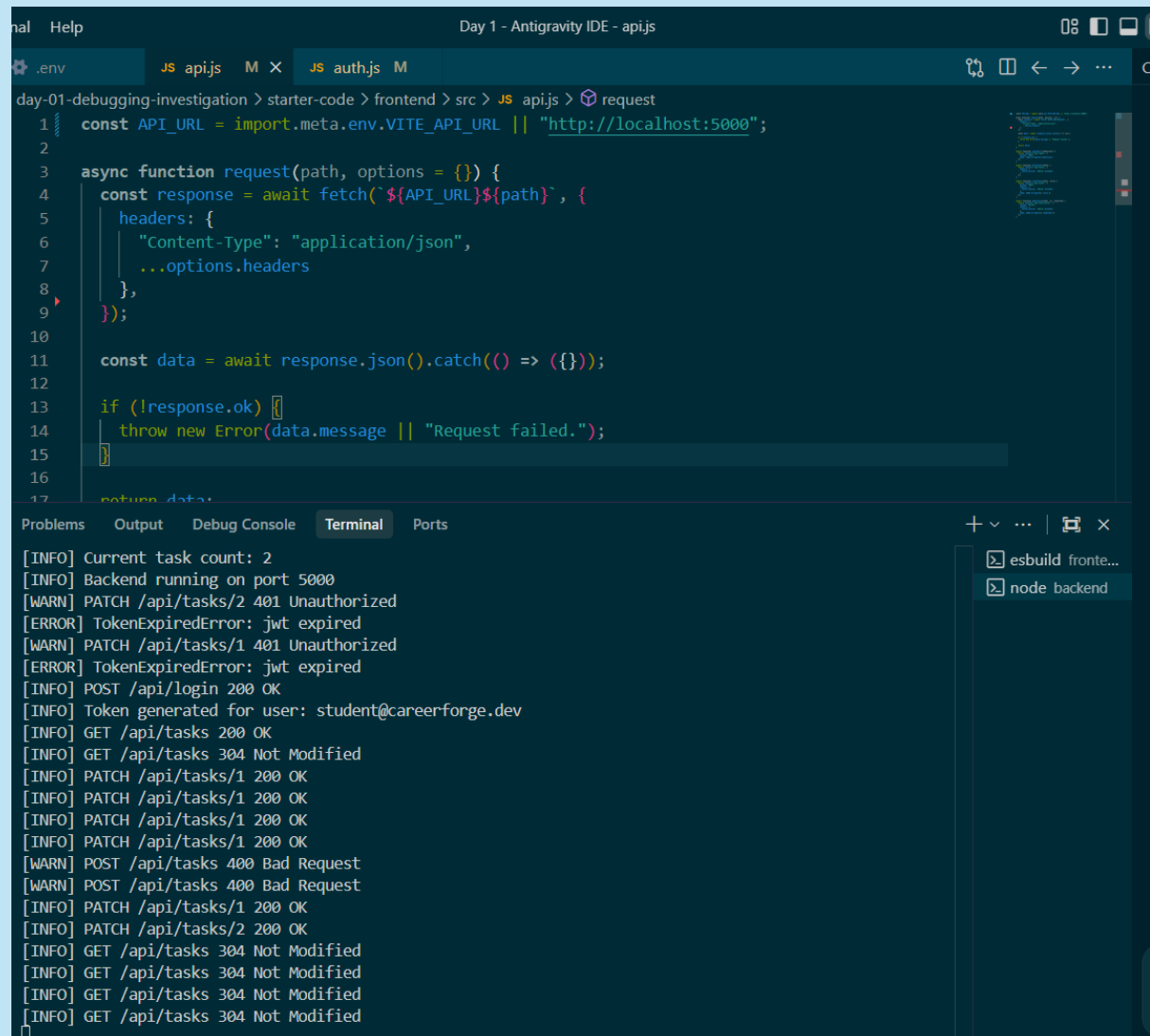
```
1  const API_URL = import.meta.env.VITE_API_URL || "http://localhost:5000";
2
3  async function request(path, options = {}) {
4    const response = await fetch(`${API_URL}${path}`, {
5      headers: {
6        "Content-Type": "application/json",
7        ...options.headers
8      },
9      ...options
10   });
11
12   const data = await response.json().catch(() => ({}));
13
14   if (!response.ok) {
15     throw new Error(data.message || "Request failed.");
16   }
17 }
```

The bottom panel contains the **Terminal** tab, showing the following logs:

```
Restarting 'src/server.js'
[INFO] In-memory task store initialized
[INFO] Current task count: 2
[INFO] Backend running on port 5000
[WARN] PATCH /api/tasks/2 401 Unauthorized
[ERROR] TokenExpiredError: jwt expired
[WARN] PATCH /api/tasks/1 401 Unauthorized
[ERROR] TokenExpiredError: jwt expired
[INFO] POST /api/login 200 OK
[INFO] Token generated for user: student@careerforge.dev
[INFO] GET /api/tasks 200 OK
[INFO] GET /api/tasks 304 Not Modified
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/1 200 OK
[WARN] POST /api/tasks 400 Bad Request
[WARN] POST /api/tasks 400 Bad Request
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/2 200 OK
```

On the right side of the terminal, there are two buttons: `esbuild fronte...` and `node backend`.

Evidence 7.2 [E7.2]



The screenshot displays the Antigravity IDE interface. The top bar shows the file path: `Day 1 - Antigravity IDE - api.js`. The editor window contains the following JavaScript code:

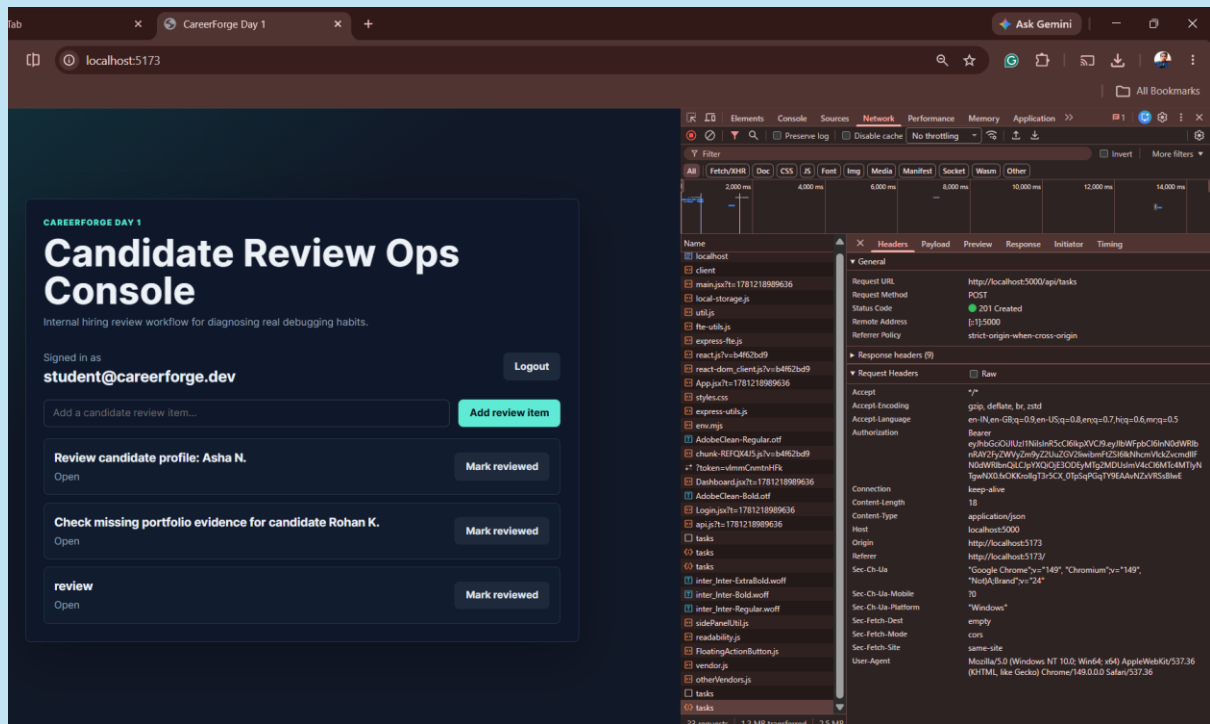
```
1  const API_URL = import.meta.env.VITE_API_URL || "http://localhost:5000";
2
3  async function request(path, options = {}) {
4    const response = await fetch(`${API_URL}${path}`, {
5      headers: {
6        "Content-Type": "application/json",
7        ...options.headers
8      },
9    });
10
11    const data = await response.json().catch(() => ({}));
12
13    if (!response.ok) {
14      throw new Error(data.message || "Request failed.");
15    }
16
17    return data;
```

The bottom panel shows the **Terminal** tab with the following logs:

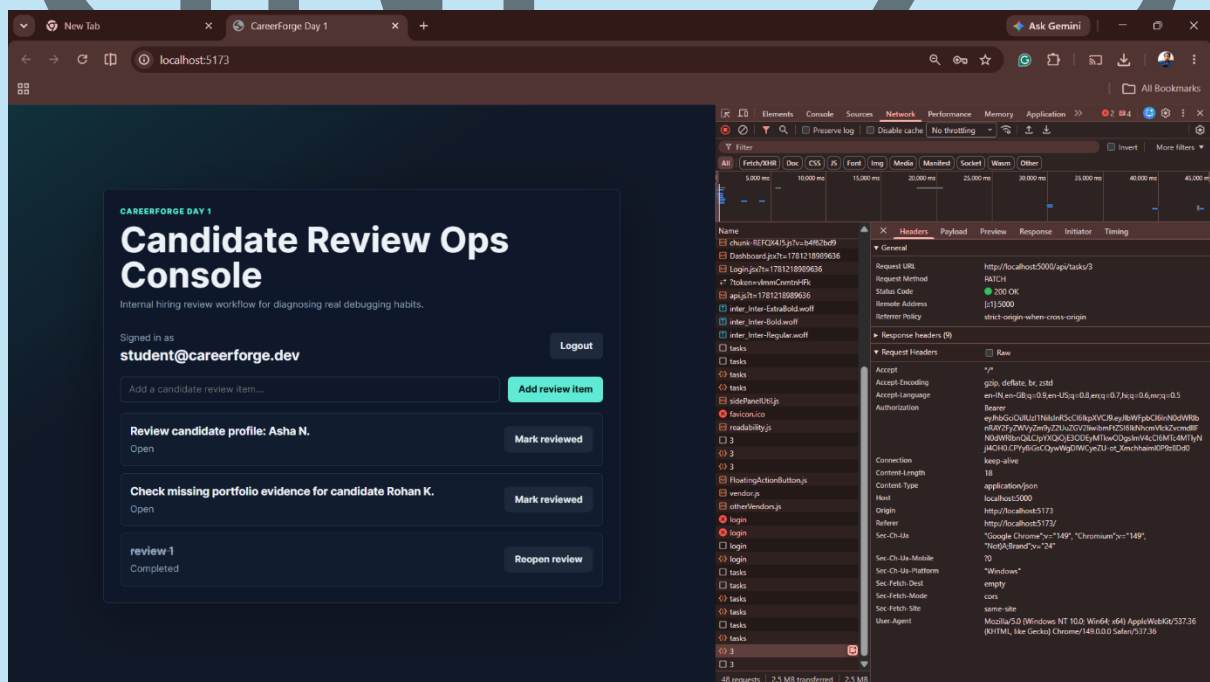
```
[INFO] Current task count: 2
[INFO] Backend running on port 5000
[WARN] PATCH /api/tasks/2 401 Unauthorized
[ERROR] TokenExpiredError: jwt expired
[WARN] PATCH /api/tasks/1 401 Unauthorized
[ERROR] TokenExpiredError: jwt expired
[INFO] POST /api/login 200 OK
[INFO] Token generated for user: student@careerforge.dev
[INFO] GET /api/tasks 200 OK
[INFO] GET /api/tasks 304 Not Modified
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/1 200 OK
[WARN] POST /api/tasks 400 Bad Request
[WARN] POST /api/tasks 400 Bad Request
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/2 200 OK
[INFO] GET /api/tasks 304 Not Modified
[INFO] GET /api/tasks 304 Not Modified
[INFO] GET /api/tasks 304 Not Modified
[INFO] GET /api/tasks 304 Not Modified
```

On the right side of the terminal, there are two tabs: `esbuild fronte...` and `node backend`.

Evidence 8 [E8]



Evidence 8.1 [E8.1]



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Engineering Investigation Report

Name: Gourik Narendra Patil

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Branch / Year: CO5KC

Repo : https://github.com/gourikpatil291-droid/careerforge7days_challenge_gnp/tree/Day-01-reality-check

Branch: day-1-reality-check

1. Failure Summary

The Candidate Review Ops Console was failing during normal reviewer operations. Login requests were being sent to an incorrect backend port, causing connection failures. After fixing the API configuration, review item creation still failed with a 400 Bad Request response. Investigation of frontend configuration, network requests, backend logs, and source code helped identify the root causes and verify the critical workflow.

2. Evidence Table

ID	Evidence Type	What I Saw	What It Proved
E1	Browser / UI	Login screen displayed errors and failed to proceed correctly	User workflow was broken
E2	Network	Login request sent to http://localhost:8000/api/login	Frontend was targeting the wrong backend
E3	Backend Log	Backend terminal showed "Backend running on port 5000"	Backend was healthy and listening on port 5000
E4	Config / Env	VITE_API_URL= http://localhost:8000 in frontend .env.example	Frontend configuration mismatch
E5	Source Code	API configuration referenced port 8000 while backend used port 5000	Root cause located in configuration
E6	Network Verification	POST /api/tasks returned 400 Bad Request	Review item creation request was failing
E7	Source Code	Authentication token generation and task request handling reviewed	Confirmed request validation behavior
E8	Backend Log	"In-memory task store initialized"	Data persistence is temporary and resets after restart
E9	Verification	Login, queue loading and review workflow worked after fixes	Fixes successfully resolved critical workflow

3. Hypothesis Trail

Observation	Hypothesis	Test Performed	Result
Login failed	Backend server is down	Checked backend terminal and health status	Rejected
Login failed	Frontend calling wrong API endpoint	Inspected Network tab and .env file	Accepted
Review item creation failed	Authentication token missing	Checked request headers	Rejected
Review item creation failed	Invalid request handling in API flow	Inspected request and response data	Accepted
Data disappeared after restart	Database connection issue	Reviewed backend startup logs	Rejected
Data disappeared after restart	Application uses in-memory storage	Inspected backend logs and behavior after restart	Accepted

4. Root Causes

Root Cause	Evidence ID	Fix Commit	Why The Fix Works
Frontend API URL pointed to port 8000 while backend ran on port 5000	E2, E3, E4, E5	fix: correct ops console API target	Requests are now sent to the active backend
Review item request handling caused 400 Bad Request errors	E6, E7	fix: correct review item request flow	API now receives valid request data
Application uses in-memory task storage	E8	verify: document persistence behavior	Behavior is understood and documented

5. Verification

Workflow	Before Fix	After Fix	Evidence ID
Backend health check	Pass	Pass	E3
Reviewer login	Failed	Pass	E9
Candidate review queue loads	Failed	Pass	E9
Review item creation	Failed with 400	Pass	E9
Review item update	Unreliable	Pass	E9
Session behavior	Functional	Functional	E9
Persistence behavior	Data lost after restart	Documented limitation	E8

6. AI Verification

AI suggested:

The frontend might be targeting the wrong backend port.

AI assumed:

The login failures were caused by an API configuration mismatch.

I verified by:

Checking the Network tab, backend logs, and frontend environment configuration.

I accepted the suggestion because:

The frontend was sending requests to localhost:8000 while the backend was running on localhost:5000. After correcting the API URL, login requests succeeded.

7. Debugging Self-Diagnosis

My weakest debugging behavior today was:

Initially assuming the login problem was caused by backend code before checking runtime evidence.

Evidence:

The backend was healthy, but the Network tab showed requests were being sent to the wrong port.

Tomorrow I need to improve:

Checking runtime evidence first before making assumptions.

Final Checklist

- I included runtime evidence.
- I included at least one rejected hypothesis.
- Every root cause connects to evidence.
- Every fix connects to a commit.
- I verified the critical workflow after fixing.
- I can defend this report live.

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